

Problem 1

Using the training and test sets specified in the syllabus, perform the following tasks:

1. On the Gisette dataset, train decision trees of maximum depth $1, 2, \dots$ up to 12, for a total of 12 decision trees. If your package does not allow the max depth as a parameter, train trees with $2^1, 2^2, \dots, 2^{12}$ nodes, again a total of 12 trees. Use the trained trees to predict the class labels on the training and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test misclassification errors vs tree depth (or $\log_2$ of nodes) as two separate curves. Report in a table the minimum test error and the tree depth (number of nodes or splits) for which the minimum was attained.
2. Repeat point 1.) on the satimage, madelon, hill-valley, and covtype datasets.
3. On the Gisette dataset, for each off $k \in \{3, 10, 30, 100, 300\}$ train a random forest with k trees where the split attribute at each node is chosen from a random subset of $71 \approx \sqrt{5000}$ features. Use the trained trrees to predict the class labels on the training and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test errors vs. number of trees k as two separate curves. Report the training and test misclassification errors in a table.
4. Repeat point 3.) on the satimage, madelon, and hill-valley datasets.
5. Repeat point 3.) on the Gisette dataset where the split attribute at each node is chosen from a random subset of $12 \approx \log_2(5000)$ features, and where the split attribute at each node is chosen from all 5000 features, plotting all four curves (2 train and 2 test misclassification errors) on the same graph.